General pathology

Chronic inflammation

Lec. 5

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Chronic inflammation

- Chronic inflammation is a response of prolonged duration (weeks or months) in which **active inflammation**, **tissue injury**, and **attempts at repair** coexist, in varying combinations .
- Chronic inflammation may follow unresolved acute inflammation or be chronic from the outset.

Causes of chronic inflammation

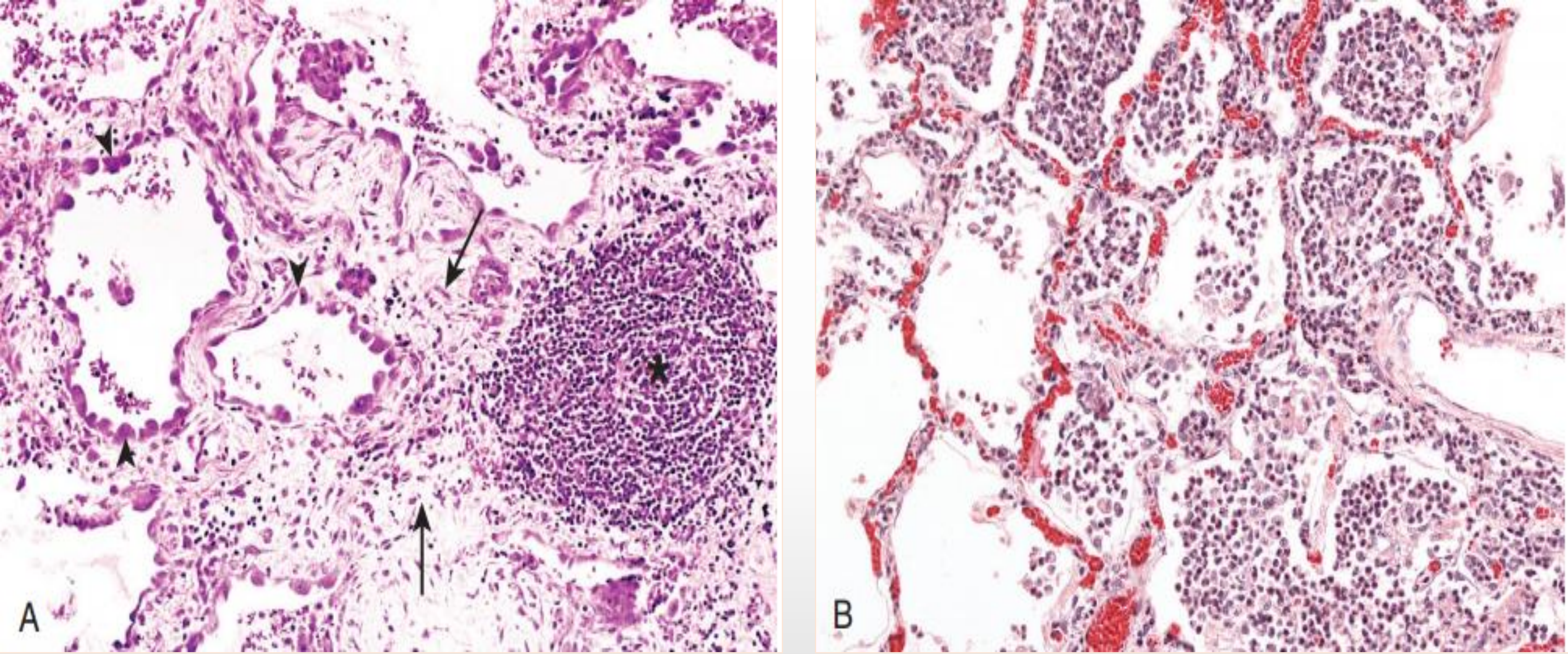
1) **Persistent infections :** by microorganisms that are difficult to eradicate, such as mycobacteria and certain viruses, fungi, and parasites.

2) **Hypersensitivity diseases** : Chronic inflammation plays an important role in a group of diseases that are caused by excessive and inappropriate activation of the immune system .

3) **Prolonged exposure to potentially toxic agents**, either exogenous or endogenous. An example of an exogenous agent is particulate silica, a nondegradable inanimate material that, when inhaled for prolonged periods, results in an inflammatory lung disease called silicosis .

Morphologic features

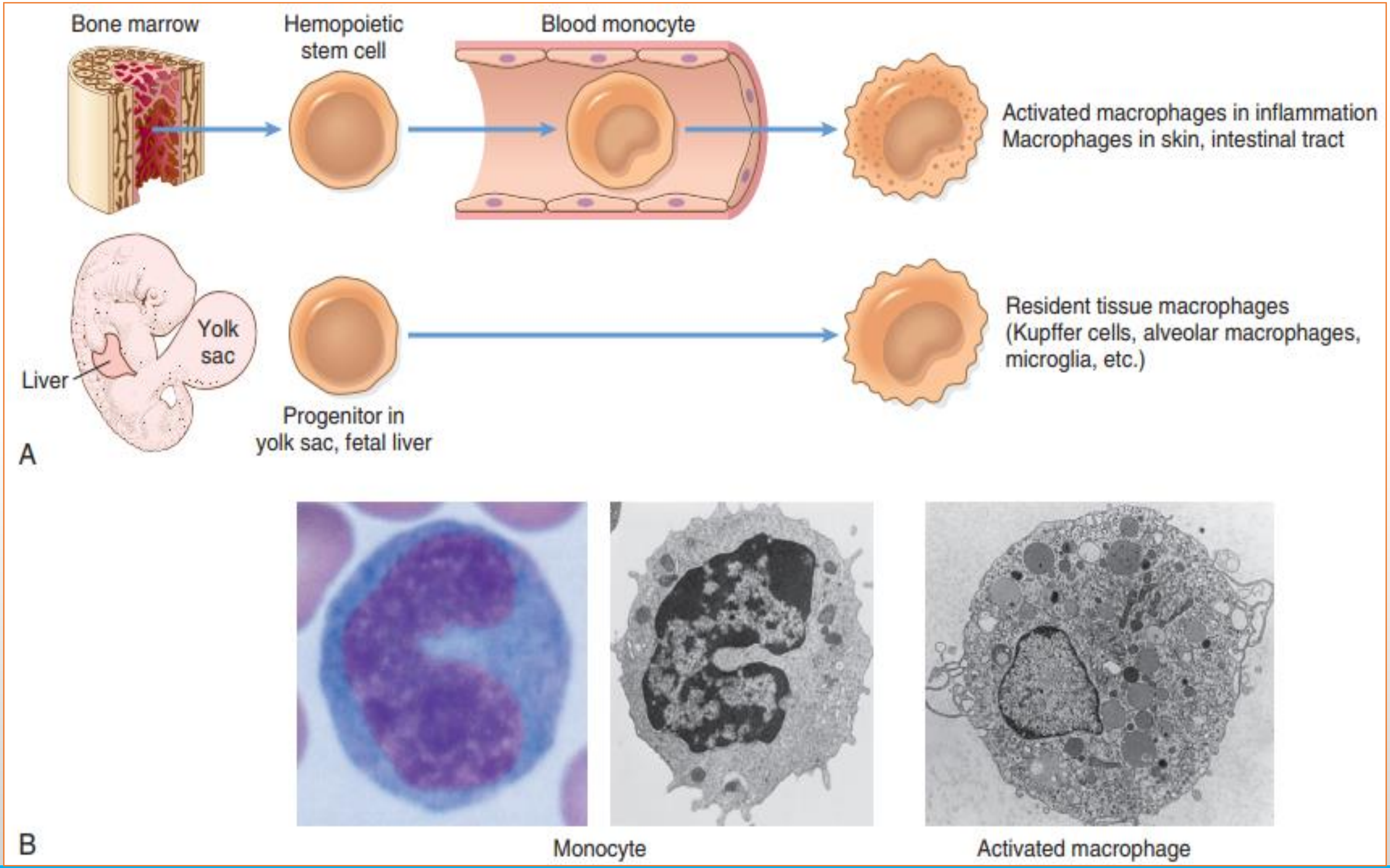
- In contrast to acute inflammation, which is manifested by vascular changes, edema, and predominantly neutrophilic infiltration, chronic inflammation is characterized by the following:
- **Infiltration with mononuclear cells**, which include macrophages, lymphocytes, and plasma cells
- **Tissue destruction**, induced by the persistent offending agent or by the inflammatory cells
- **Attempts at healing** by connective tissue replacement of damaged tissue, accomplished by angiogenesis (proliferation of small blood vessels) and, in particular, fibrosis .



Chronic inflammation in the lung, showing all three characteristic histologic features: (1) collection of chronic inflammatory cells (*), (2) destruction of parenchyma (normal alveoli are replaced by spaces lined by cuboidal epithelium, arrowheads), and (3) replacement by connective tissue (fibrosis, arrows). (B) In contrast, in acute inflammation of the lung (acute bronchopneumonia), neutrophils fill the alveolar spaces and blood vessels are congested.

Chronic inflammatory cells

- **Macrophages**
- They are the dominant cells in most chronic inflammatory reactions . Macrophages are tissue cells derived from hematopoietic stem cells in the bone marrow and from progenitors in the embryonic yolk sac and fetal liver during early development . Circulating cells of this lineage are known as **monocytes.** Macrophages are normally diffusely scattered in most connective tissues.


Bone marrow
Hemopoietic stem cell
Blood monocyte
Activated macrophages in inflammation
Macrophages in skin, intestinal tract
Yolk sac
Liver
Progenitor in yolk sac, fetal liver
Resident tissue macrophages (Kupffer cells, alveolar macrophages, microglia, etc.)
A
B
Monocyte
Activated macrophage

- **Macrophages are also found in specific locations in organs such as**
- Liver (called Kupffer cells)
- Spleen and lymph nodes (sinus histiocytes)
- Central nervous system (microglial cells)
- Lungs (alveolar macrophages).

- **Functions of macrophages**

1) Professional phagocytes that act as filters for particulate matter, microbes, and senescent cells .

2) Initiate the process of tissue repair and involved in scar formation (growth factor)

3) Secrete mediators of inflammation such as (Tumor necrotic factor TNF, Interleukin -1 , and Cytokines) to promote leukocyte recruitment.

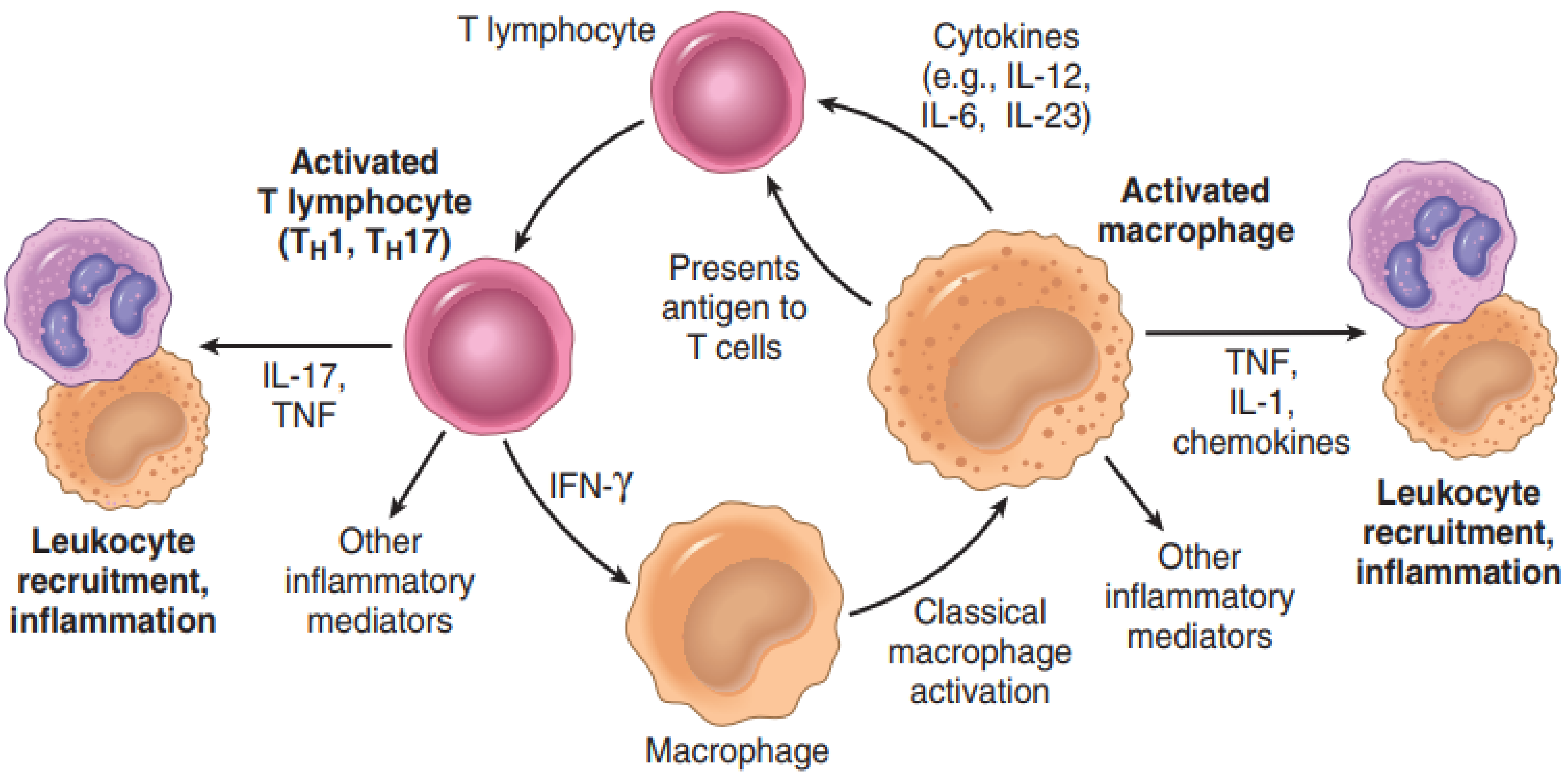
4) Display antigens to T lymphocyte and respond to signal from T cell.

- **Lymphocytes**

- Both T and B lymphocytes migrate into inflammatory sites using the same chemical mediator that recruit other leukocytes.

- Lymphocytes and macrophages interact in a **bidirectional way**, and these interactions play an important role in chronic inflammation. Macrophages produce cytokines (IL-12) that stimulate T-cell responses.

- Activated T lymphocytes, in turn, produce IFN-γ, is a powerful activator of macrophages, promoting more antigen presentation and cytokine secretion. The result is a cycle of cellular reactions that fuel and sustain chronic inflammation.



Macrophage–lymphocyte interactions in chronic inflammation. Activated T cells produce cytokines that recruit macrophages (TNF, IL-17, chemokines) and others that activate macrophages (IFN-γ). Activated macrophages in turn stimulate T cells by presenting antigens and via cytokines such as IL-12.

- **Plasma cells** : are activated B-lymphocyte that can produce immunoglobulin molecules.
- **Eosinophils** : are abundant in immune reactions such as allergies and in parasitic infections . Eosinophils have granules that contain major basic protein, a highly cationic protein that is toxic to parasites but also injures host epithelial cells.
- **Mast cells :** are widely distributed in connective tissues and participate in both acute and chronic inflammatory reactions

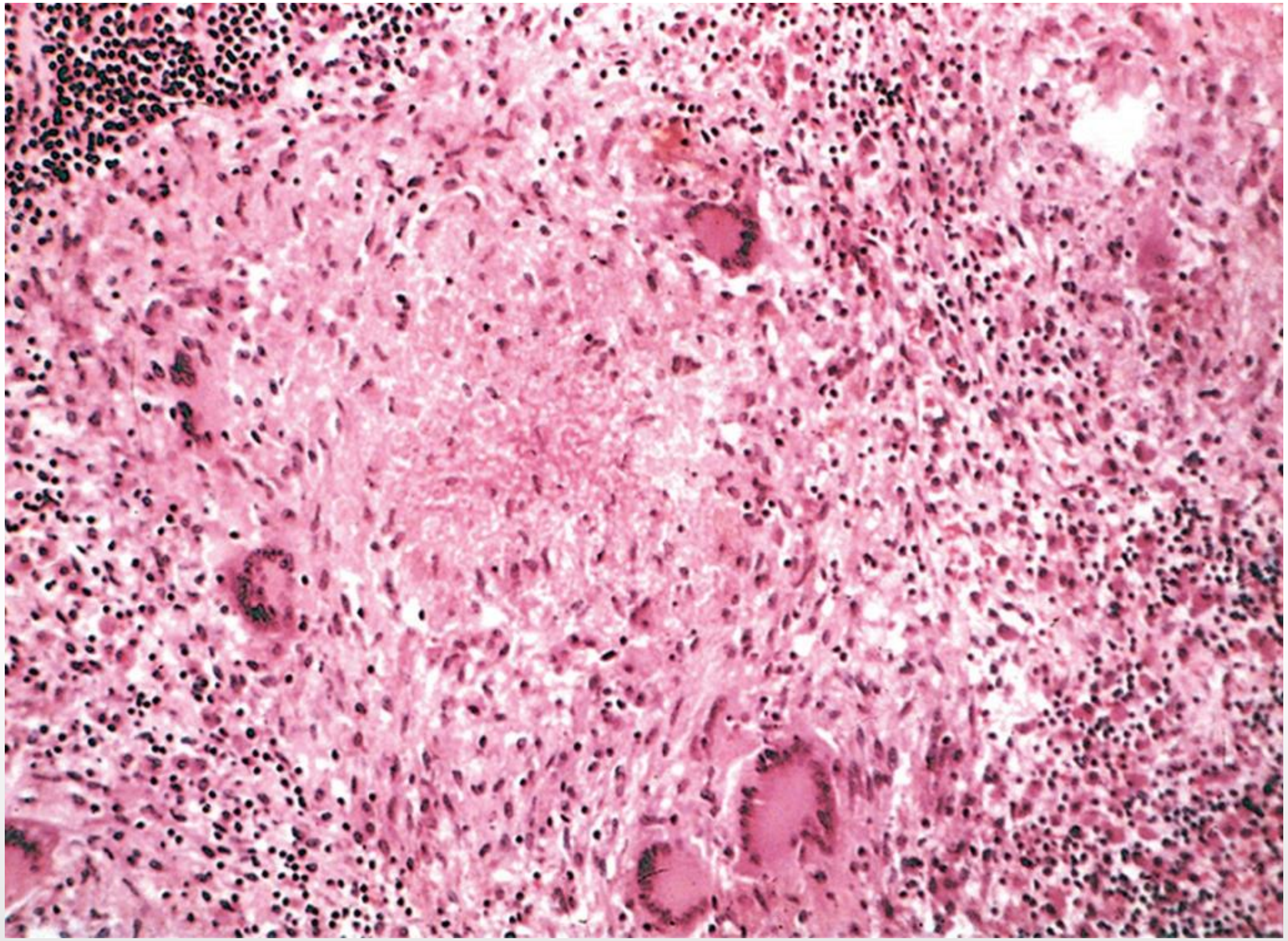
Granulomatous Inflammation

- Granulomatous inflammation is a form of chronic inflammation characterized by collections of activated macrophages, often with T lymphocytes, and sometimes associated with central necrosis.

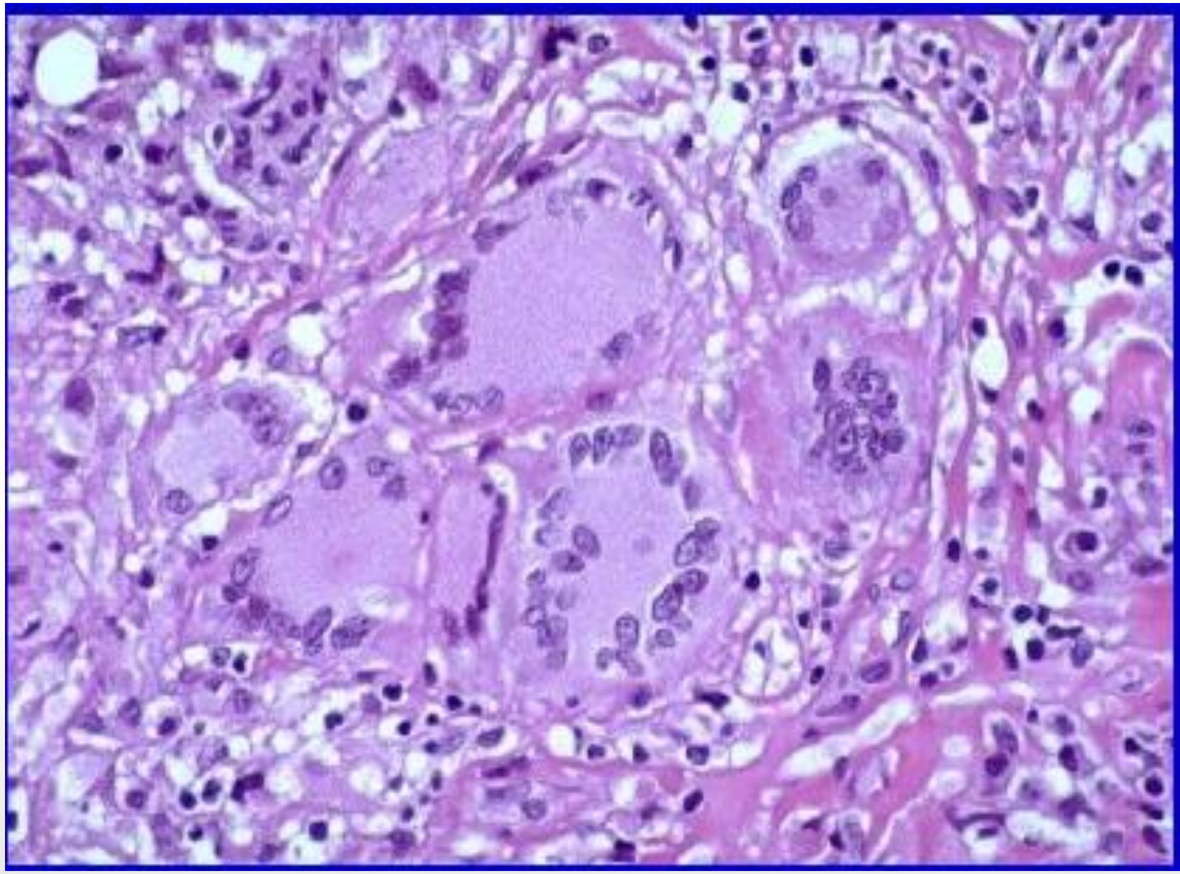
- **Granuloma** formation is a cellular attempt to contain an offending agent that is difficult to eradicate. In this attempt there is often strong activation of T lymphocytes leading to macrophage activation, which can cause injury to normal tissues.

Morphology

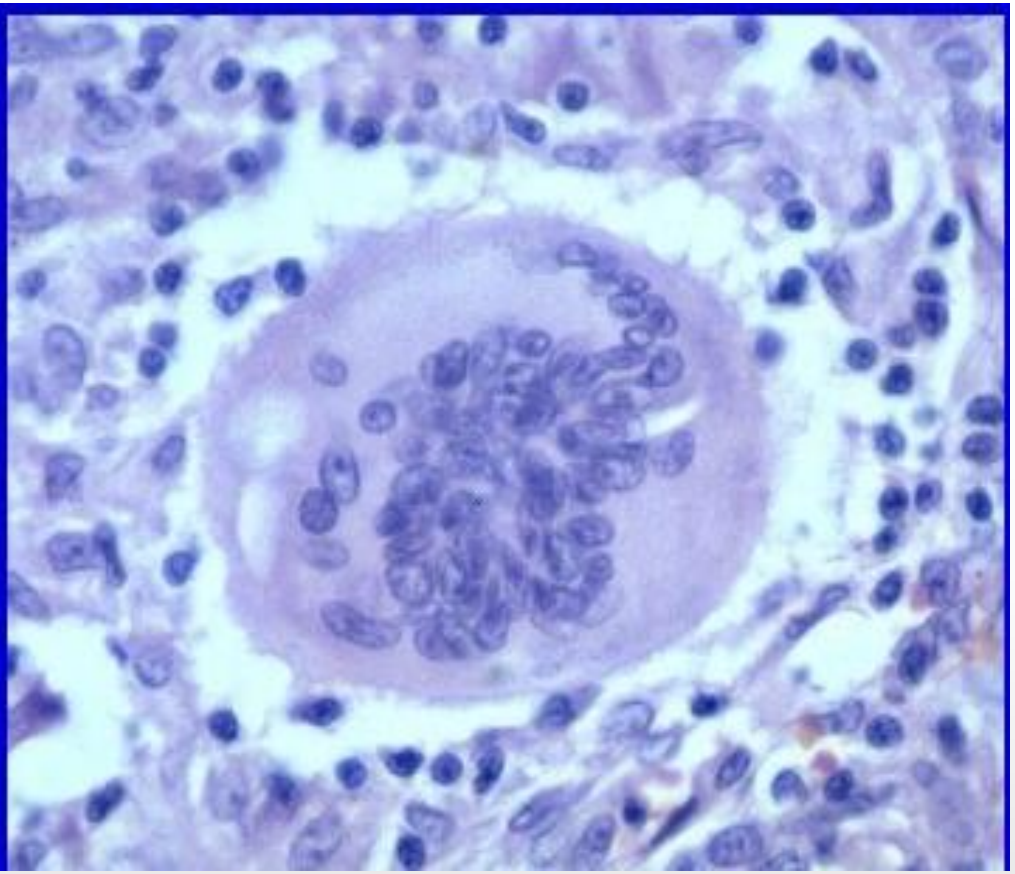
- **Epithelioid Cells**: activated macrophages may develop abundant cytoplasm and begin to resemble epithelial cells .
- **Multinucleated giant cells :** derive from the fusion of multiple activated macrophages and consist of a large mass of cytoplasm with many nuclei(20 or more) arranged either peripherally Langhans-type giant cell or haphazardly (foreign body-type giant cell) .
- Granulomas associated with certain infectious organisms (classically Mycobacterium tuberculosis) often contain a **central zone of necrosis.**



Typical tuberculous granuloma showing an area of central necrosis surrounded by multiple multinucleate giant cells, epithelioid cells, and lymphocytes



Langhans-type giant cell



Foreign body-type giant cell